MIL-E-1/520B
26 November 1968
SUPERSEDING
MIL-E-1/520A
27 May 1958

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPES 6J7 AND 12J7GT [1/]
a b

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, RF sharp cutoff

Outline --- a--- 8-4(EIA)
b--- 9-18(EIA)

Base --- a--- B7-22
b--- B7-27

Envelope --- a--- MTT8
b--- T-9

Cathode --- a,b- Coated unipotential

Base connections:

Pin No.	---	*a,b*	*1*	*2*	*3*	*4*	*5*	*7*	*8*	*Cap*
Element	---	a	sh	h	a	g2	g3	h	k	g1
		b	base sleeve	h	a	g2	g3	h	k	g1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ef	Eb	Ec1	Ec2	Ec3	Pp	Pg2	Ehk	Alt
Unit:		Vdc	Vdc	Vdc	Vdc	Vdc	W	W	v	ft
Maximum:	a	6.9	330	---	140	0	0.8	0.11	100	10,000
	b	13.8	330	---	140	0	0.8	0.11	100	10,000
Minimum:	a	5.7	---	---	---	---	---	---	---	---
	b	11.4	---	---	---	---	---	---	---	---
TEST CONDITIONS:	a	6.3	250	-3.0	100	0	---	---	---	---
	b	12.6	250	-3.0	100	0	---	---	---	---

GENERAL:

Qualification - Not required

1/ See note 2

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1266	Total grid current	a, b	See note 1	0.65	II	Ic1	0	-0.5	μAdc
1256	Electrode current (1) (anode)	a, b		0.65	II	Ib	1.3	2.9	mAdc
1256	Electrode current (screen-grid)	a, b		0.65	II	Ic2	0.2	1.0	mAdc
1306	Tranconductance	a, b		0.65	II	Sm	1,000	1,450	μmhos
1231	Emission current	a, b	Eb = Ec1 = Ec2 = Ec3 = 30 Vdc; (see note 1)	0.65	II	Is	45	---	mAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Heater current	a		6.5	S3	If	275	325	mA
		b		6.5	S3	If	138	162	mA
1031	Low-frequency vibration	a, b	Rp = 10,000 ohms	6.5	S3	Ep	---	200	mVac
Ⓑ1336	Heater-cathode leakage	a, b		6.5	S3	Ihk	---	20	μAdc
1256	Electrode current (2) (anode)	a, b	Ec1 = -8 Vdc	6.5	S3	Ib	0	30	μAdc
1211	Insulation of electrodes	a, b		4.0	S3	R	10	---	Meg
1331	Direct-interelectrode capacitance	all	Shield No. 308; T-9 bulb only	6.5	S3	Cgp	---	0.005	pF
		a	Without shield			Cin	5.6	8.4	pF
		b	Shield No. 308			Cin	3.6	6.2	pF
		all	Shield No. 308; T-9 bulb only			Cout	8.4	15.6	pF
1101	Secureness of base and base insert	a, b		6.5	S3	---	---	---	---
1111	Base pin solder depth (rigid leads)	a, b		6.5	S3	---	---	---	---
1131	Metallic base sleeve quality	b		6.5	S3	---	---	---	---
1105	Permanence of marking	a, b		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3								
1501	Intermittent life	a, b	Group A; Ehk = 100 V	---	---	---	---	---	---
---	Life-test end point (intermittent) (500 hours)	a, b	Transconductance	---	---	Sm	825	---	µmhos

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Tube type 6J7GT is hereby deleted from this TSS, Tubes for replacement purposes will be procured to EIA data, Release Number 134.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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